BarrierLab: Mathematical Specification

Measurement version `shared-outcome-cache-float64-v2`

Contents

1 Scope

This document specifies the numerical procedure implemented by the three-stage BarrierLab pipeline. Caching and artifact references change where a tensor is obtained, but they do not change any equation below.

2 Notation and tensor axes

Let

- $p \in \{1, \ldots, P\}$ identify a price history;
- $i \in \{1, \ldots, N\}$ identify an observation date;

- $t \in \mathcal{T}$ identify a forward horizon;
- $\delta \in \mathcal{D}$ identify a signed return barrier;
- $b \in \{1, \ldots, B\}$ identify a condition bin; and
- $x_{p,i}$ denote a condition-feature value.

The default NASDAQ grid currently has $|\mathcal{D}| = 41$, $|\mathcal{T}| = 30$, and requests $B = 10$ quantile bins. Stage 3 currently uses $P = 1{,}000$ null histories.

Table 1: Principal tensor axes. An observed history has $P = 1$.

Quantity	Axes
OHLCV histories	$P \times N \times 5$
Forward low/high excursions, one horizon	$P \times N$
Barrier touches, one horizon	$P \times N \times |\mathcal{D}|$
Conditional probabilities	$P \times |\mathcal{D}| \times B \times |\mathcal{T}|$
Baseline probabilities	$P \times |\mathcal{D}| \times |\mathcal{T}|$
Bin scores	$P \times B$

Synthetic histories are evaluated in memory-bounded path batches. The batch axis is mathematically part of P and does not alter the statistic.

3 Forward price excursions

For every path, starting date, and horizon, define the greatest downside and upside excursion reached by future intrabar prices:

$$L_{p,i,t} = \frac{\min_{1 \le k \le t} \mathrm{Low}_{p,i+k}}{\mathrm{Close}_{p,i}} - 1, \tag{1}$$

$$H_{p,i,t} = \frac{\max_{1 \le k \le t} \mathrm{High}_{p,i+k}}{\mathrm{Close}_{p,i}} - 1. \tag{2}$$

Dates without t future observations are non-finite and therefore ineligible. Horizons are traversed incrementally: the running future minimum and maximum for horizon t extend the state calculated for shorter horizons.

4 Signed-delta touch matrix

All barriers are broadcast over the excursion tensor. Define

$$T_{p,i,t,\delta} = \begin{cases} \mathbf{1}[L_{p,i,t} \le \delta], & \delta < 0, \\ \mathbf{1}[H_{p,i,t} \ge \delta], & \delta \ge 0. \end{cases} \tag{3}$$

For one path batch and horizon, T is a Boolean $P \times N \times |\mathcal{D}|$ matrix. It depends only on prices, horizon, and the delta grid; it does not depend on a feature, node, or condition bin. The implementation constructs it once and reuses it for the baseline and every node. The zero barrier belongs to the measured probability surface but is excluded from the paired-barrier score in Section 9.

5 Feature bins

For a continuous feature, the requested interior quantile edges are

$$e_j = Q_{j/B}(\{x_i : x_i \text{ is finite}\}), \qquad j = 1, \dots, B-1. \tag{4}$$

Non-finite and duplicate edges are removed. Edges outside $\min(x) < e_j \leq \max(x)$ are also removed. Heavy ties can therefore produce fewer than B effective bins.

Bin assignment uses left insertion:

$$b_i = \mathrm{searchsorted}(e, x_i, \mathrm{left}). \tag{5}$$

Consequently, an observation equal to an edge enters the lower bin. Stage 1 stores feature values, edges, and the resulting integer bin assignments.

For a synthetic OHLCV-derived feature, the feature and its quantile edges are calculated independently for every synthetic path. External, calendar, and workspace-extension conditions that cannot be reconstructed from synthetic OHLCV use their observed feature values and observed edges for every null path.

6 Conditional and unconditional probabilities

Define price and condition eligibility by

$$E^{\text{price}}_{p,i,t} = \mathbf{1}[L_{p,i,t}, H_{p,i,t} \text{ are finite}], \tag{6}$$

$$E^{\text{condition}}_{p,i,t} = E^{\text{price}}_{p,i,t}\mathbf{1}[x_{p,i} \text{ is finite}]. \tag{7}$$

For condition bin b, horizon t, and barrier δ, define

$$n_{p,b,t} = \sum_i E^{\text{condition}}_{p,i,t}\mathbf{1}[b_{p,i} = b], \tag{8}$$

$$h_{p,\delta,b,t} = \sum_i T_{p,i,t,\delta}E^{\text{condition}}_{p,i,t}\mathbf{1}[b_{p,i} = b], \tag{9}$$

$$C_{p,\delta,b,t} = \frac{h_{p,\delta,b,t}}{n_{p,b,t}}. \tag{10}$$

The conditional probability $C_{p,\delta,b,t}$ is non-finite when $n_{p,b,t} < 30$.

The unconditional market baseline includes every price-eligible date and is independent of feature warm-up:

$$n^{\text{base}}_{p,t} = \sum_i E^{\text{price}}_{p,i,t}, \tag{11}$$

$$B_{p,\delta,t} = \frac{\sum_i T_{p,i,t,\delta}E^{\text{price}}_{p,i,t}}{n^{\text{base}}_{p,t}}. \tag{12}$$

The baseline is non-finite when $n^{\text{base}}_{p,t} < 30$. Every synthetic path uses its own baseline; null histories are never compared with the observed market's baseline.

7 Stage 1: measurement

Stage 1 measures the complete conditional surface $C_{\delta,b,t}$, hit counts $h_{\delta,b,t}$, observation counts $n_{b,t}$, and the unconditional surface $B_{\delta,t}$. Observed forward excursions, touches, and baseline probabilities are retained as float64 source-level cache tensors. Feature values, quantile edges, and bin assignments are retained in the node artifact.

8 Stage 2: baseline-relative shift

Stage 2 calculates the percentage-point deviation from the market baseline:

$$S_{p,\delta,b,t} = 100\,(C_{p,\delta,b,t} - B_{p,\delta,t})\,. \tag{13}$$

It stores only S plus cryptographic references to the node and baseline Stage 1 artifacts. Probabilities, counts, axes, conditions, and market history are materialized from those referenced artifacts when required.

9 Full-grid bin score

Let $\mathcal{M}$ be the set of nonzero magnitudes d for which the delta grid contains exactly one $+d$ barrier and one $-d$ barrier. For each pair, bin, and horizon, define the signed-barrier asymmetry

$$A_{p,d,b,t} = |S_{p,+d,b,t} - S_{p,-d,b,t}|\,. \tag{14}$$

Each magnitude receives the linear weight

$$w_d = \frac{d}{\max_{m\in\mathcal{M}} m}. \tag{15}$$

A contribution is usable only if

1. its positive and negative shifts are finite;
2. its bin contains at least 30 eligible observations; and
3. at least two bins are usable for that magnitude and horizon.

Let $U_{p,d,b,t}$ be the corresponding Boolean usability indicator. The full score for a condition bin is

$$Q_{p,b} = \sum_{t\in\mathcal{T}} \sum_{d\in\mathcal{M}} U_{p,d,b,t}\, w_d\, A_{p,d,b,t}. \tag{16}$$

An unsupported bin has score zero and a separate false-validity flag. A supported bin whose contributions genuinely sum to zero remains valid.

10 Synthetic null model

For the observed history, define the four per-bar log components

$$r_i^{\text{open}} = \log\!\left(\frac{\text{Open}_i}{\text{Close}_{i-1}}\right), \qquad r_i^{\text{close}} = \log\!\left(\frac{\text{Close}_i}{\text{Open}_i}\right), \tag{17}$$

$$r_i^{\text{high}} = \log\!\left(\frac{\text{High}_i}{\max(\text{Open}_i, \text{Close}_i)}\right), \qquad r_i^{\text{low}} = \log\!\left(\frac{\text{Low}_i}{\min(\text{Open}_i, \text{Close}_i)}\right). \tag{18}$$

After non-finite rows are removed, let μ and Σ be the empirical mean vector and covariance matrix. A small diagonal jitter is added before Cholesky factorization:

$$\epsilon = 10^{-12} \max\!\left(1, \max_j |\Sigma_{jj}|\right), \tag{19}$$

$$F = \operatorname{chol}(\Sigma + \epsilon I). \tag{20}$$

Using deterministic seed `20260907`, each null innovation is

$$z_{p,i} \sim \mathcal{N}(0, I_4), \qquad r_{p,i} = z_{p,i} F^{\mathsf{T}} + \mu. \tag{21}$$

The draws preserve contemporaneous covariance among the four OHLC components, but they are independent across time.

Synthetic prices are reconstructed by

$$\mathrm{Close}_{p,i} = \mathrm{Close}_0 \exp\left(\sum_{j\le i}(r_{p,j}^{\mathrm{open}} + r_{p,j}^{\mathrm{close}})\right), \tag{22}$$

$$\mathrm{Open}_{p,i} = \mathrm{Close}_{p,i-1} \exp(r_{p,i}^{\mathrm{open}}), \tag{23}$$

$$\mathrm{High}_{p,i} = \max(\mathrm{Open}_{p,i}, \mathrm{Close}_{p,i}) \exp(r_{p,i}^{\mathrm{high}}), \tag{24}$$

$$\mathrm{Low}_{p,i} = \min(\mathrm{Open}_{p,i}, \mathrm{Close}_{p,i}) \exp(r_{p,i}^{\mathrm{low}}). \tag{25}$$

Observed volume is broadcast across synthetic histories. The null does not model temporal autocorrelation, volatility clustering, regime transitions, liquidity, execution, or trading costs.

11 Null scoring and hypothesis test

Every synthetic history passes through the same excursion, touch, conditional probability, baseline-shift, and bin-score equations used for the observation. Within a path batch and horizon, the touch matrix is generated once; baseline reduction and every node-specific bin reduction consume that same matrix.

For observed score Q_b^{obs} and P null scores $Q_{1,b}^{\mathrm{null}}, \ldots, Q_{P,b}^{\mathrm{null}}$, the one-sided Monte Carlo p-value is

$$p_b = \frac{1 + \sum_{p=1}^{P} \mathbf{1}\left[Q_{p,b}^{\mathrm{null}} \ge Q_b^{\mathrm{obs}}\right]}{P+1}. \tag{26}$$

With the current $P = 1{,}000$, the denominator is 1,001. A bin clears the current raw threshold precisely when

$$p_b < 0.05. \tag{27}$$

Invalid null bins retain zero scores in the complete null ensemble. The output separately records the number of null paths that produced a valid score. The current decision rule applies no family-wise or false-discovery-rate correction across bins.

12 Stage 4 selection

Stage 4 introduces no new statistical score. Let c_b be the Stage 3 `cleared` indicator for condition bin b. The selected set is exactly

$$\mathcal{S} = \{b : c_b = \mathrm{true}\} = \{b : p_b < 0.05\}. \tag{28}$$

No top-k truncation, family quota, effect-size gate, or representative-cell rule is applied. Ordering by ascending p_b and descending Q_b^{obs} is presentational only. For every $b \in \mathcal{S}$, Stage 4 renders the complete surface $S_{\delta,b,t}$ as a signed-barrier-by-horizon heatmap.

13 Cached mathematical identities

The following quantities are reused exactly rather than approximated:

- observed forward-excursion tensors;
- observed signed-delta touch matrices;
- observed unconditional baseline probabilities;
- observed feature values, quantile edges, and bin assignments;
- one synthetic touch matrix across the baseline and every node in a batch;
- common rolling feature primitives within a history or synthetic batch.

Observed caches are keyed by ordered market-history content and validated against the measurement version, delta grid, and horizon grid. Stage 2 references are protected by SHA-256 hashes. A changed history, grid, numerical version, or referenced Stage 1 artifact invalidates reuse.